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Techno-economic analysis of a microgrid design for a commercial health facility in Ghana- Case study of Zipline Sefwi-Wiawso

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A B S T R A C T

Most commercial entities depend on the availability, and reliability of electricity supply for their business ventures. Poor energy security, unreliability, and high cost of electricity characterize the utility grid of Ghana, forcing most of these facilities to resort to using diesel generators to supplement their energy needs. This study, therefore, proposes the development of a Microgrid (MG) to provide electricity to the Zipline facility in Sefwi-Wiawso, Ghana. The optimally designed MG is achieved using the HOMER software and consists of 94.8 kW Sunpower solar panels, 231 kWh Samsung M8068 Li-ion batteries, 56.31 kW ABB MGS inverter, 158 kW CAT-C7 diesel generator, and an infinite distribution grid. The proposed design with a lifespan of 25 years has an initial capital cost of US$ 45,929.17 and a nominal discounted payback period of 2.11yrs making a total savings of US$ 322,563.21.

Introduction

Access to secured and reliable electrical energy improves the socioeconomic development of every country. It enhances innovation and investment in new business ventures, which is a tool for job creation, prosperity, and the social well-being of citizens [1–3]. Electricity availability has therefore become a key component in determining the viability of most commercial activities. Many commercial settings across the world utilize high amounts of electricity and it constitutes about 61% of their total energy use [4]. Conventional power generating stations are normally situated at far locations from load centres [5]. This necessitates the construction of long Transmission and Distribution (T&D) lines to serve the required load. Constructing these long T&D networks are costly and subsequently forms part of the overall customers tariff [6].

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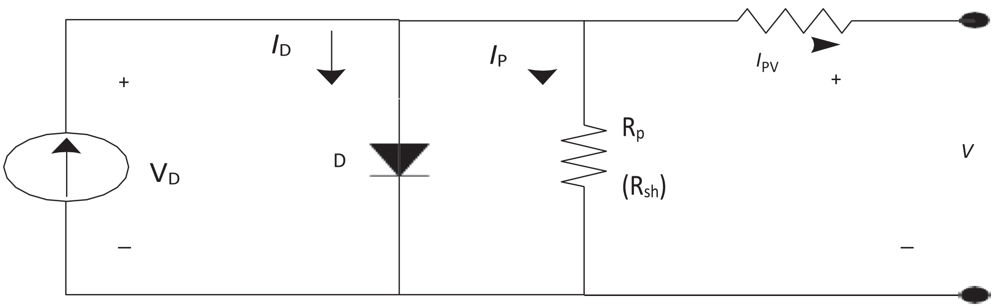


Fig. 1. Single-diode model of solar cell equivalent circuit [17,19].

The current trend is shifting from utilizing fossil fuels in conventional power plants to more sustainable and environmentally friendly fuels [7,8]. MGs are developed as power systems consisting of systems typically less than 100 kW. These systems can be Micro Hydro, Wind Turbine generators, Solar Photo Voltaic (PV), Biofuels, Fuel Cells, Geothermal energy, and Tidal Waves, among others [9]. MGs are reliable, secure, efficient, and a cleaner way of providing electricity to customers. The proximity of renewable energy sources to load centers however, necessitate evaluating the technical feasibility and economic viability of MG projects for specified areas [10].

Ghana has in the last decade lost an average of US$ 2.1 million in daily production due to high T&D losses and over-reliance on conventional hydro and thermal power plants [10,11]. Even though electricity demand is lower than installed capacity, high distribution losses have led to an escalating electricity tariff in Ghana [12]. The problem of unreliable supply [13] causing most business facilities to resort to falling on backup generators. A constant rise of fuel prices experienced in the country [14] make the utilization of such generators also expensive. These issues eventually reflect in the high cost of providing goods and services in Ghana.

A research conducted on MG design in Ghana either focused on rural electrification or commercial electricity [15,16] highlight that the desired component selection for an MG is dependent on the type of design. While there has been proportional research on microgrid design in Ghana, there has been very limited research on a design for health facility in Ghana. This study is an attempt to fulfill that gap by focusing on MG design for a health facility in Ghana.

The paper is organised as follows; section 2 provides the available literature on MG's. Section 3 follows with materials and methods. Section 4 describes the results and discussion. Finally, the conclusion and remarks are offered in section 5.

Literature review

Developing solar photovoltaic systems for electricity production

Solar energy is derived from photovoltaic systems using solar panels. The predominant type of solar panel is the silicon, polycrystalline and monocrystalline thin films. Solar cells are made by fusing N-type and P-type layers of semiconductors which create a built-in electric field. These panels use the photoelectric effect to convert energy from the sun into electricity. The energy hitting and absorbed by these panels assists the available electrons in their atoms to gain extra energy. The electrons get dissociated and eventually lead to the flow of electricity in the solar panels [17,18].

A solar cell is the basic structure of a solar PV system, interconnecting several cells to form a solar PV module. The output voltage of the solar module is increased by connecting the solar cells in series. Connecting more modules in parallel yields a higher system output current. These interconnections produce solar PV arrays. The arrays are normally designed based on the desired output voltage and current [17,19]. The current-voltage (I-V) characteristics of a solar PV array are non-linear. Eq. (1) is the output voltage equation of a silicon-based solar PV model [19,20].

$$V_{PV} = \frac{N_s \alpha kT}{q} \mathrm{In}\left[\frac{I_{SC} - I_{PV} + N_P I_0}{N_P I_0}\right] - \frac{N_s}{N_P} R_S I_{PV} \tag{1}$$

Where N_s is the number of cells in series per string where;

α is the ideality factor, k is the Boltzmann's constant (J/□K), T is the PV cell temperature □K, q is the electron charge in coulomb, I_{sc} is the short circuit current, I_{PV} PV cell output current, N_p is the number of cells per string, I_0 is Cell reverse saturation current, R_S is the Cell series resistance.

Considering the open-circuit voltage and short-circuit current, the equivalent model of a single-diode solar PV cell can be expressed in Fig. 1. This can be achieved by utilizing Eqs. (1) to 4 with V_T as the temperature of the solar device [21].

$$I_D = I_0\left[e^{V_{PV}/\alpha V_T} - 1\right] \tag{2}$$

Where, I_D is the diode current

$$I_{PV} = I_{SC} - I_D \tag{3}$$

$$V_{PV} = \alpha V_T \ln\left[\frac{I_{SC}-I_{PV}}{I_0}+1\right] \quad (4)$$

The illumination provided by the sun determines the short circuit current of the solar system, whereas the type of material used for the solar panel and the temperature (V_T) realised when the sun hits the panel determines the open-circuit voltage [21,22].

For the PV system to meet the grid requirement, conditioning of the incoming power from the solar PV must be done. A typical solar system generates DC power, nevertheless, the grid system is normally AC. For Maximum Power Point Tracking (MPPT) of a solar array, two solar system topologies are employed. That is the single inverter and double inverter topology [17,23,24]. A single AC to DC power inverter can be employed to convert the DC power from the solar system to the AC bus of the grid system. This topology is cheaper, simple, and more efficient since the MPPT and control are achieved in a single stage [25,26].

Empirical review on microgrid designs

MGs have gained prominence due to the economic, technological, and environmental effects posed by existing grid systems [27,28]. They tend to be autonomous hence any disturbance occurring on a long transmission system does not affect an MG's capability of assuming autonomy.

In Ghana, MG research has gained prominence in recent times. [29] explored the role of MG in low energy transition in Sub–Saharan Africa (SSA) with northern Ghana as a case study. The paper found that despite adequate investment in MG from the World Bank for various MG projects, the region is yet to realize adequate progress to achieve universal access to all by 2030.

According to [30] the barriers to accelerating the deployment of distributed renewable energy MG in Ghana were predominantly political barriers (44,3%) with environmental barriers (6.4%) being the least. [30] adopted the hierarchy process (AHP) method. The research recommended a modification of the policy decision with adequate investment to achieve universal access.

Furthermore, a feasibility analysis of an off-grid MG hybrid energy system for rural electrification in the Northern region of Ghana was conducted by [31]. This research studies the techno-economic analysis of a PV, with a diesel generator and battery storage hybrid power system using HOMER software package for the simulation analysis. Finding reveals that, even though the COE is almost three times the energy cost lately in Ghana, the sensitivity analysis demonstrates that varying certain constraints for example fuel, and capital subsidies can reduce COE.

Ramde et al. [32] argues that it would, therefore, be beneficial first to implement a solar MG system only from the beginning and then later invest in a battery energy storage system (BESS). This was revealed in their paper on the planning of sustainable and stable MGs for Ghanaian hospitals with photovoltaics. [32] posits a most profitable system solution by adding a 150 kWp PV plant with no BESS. Contrarily, [33] used HOMER to examine the techno-economics of solar PV–diesel hybrid power systems for off–grid outdoor base transceiver stations in Ghana. The findings indicate that coupling MG with a diesel generator and a solar PV diesel hybrid system could reduce the LCOE by 48% and cut down on emissions by up to 90%.

Nevertheless, [34] confirmed MG system yielded a very good payback period of 10.43 years about a lifetime of 25 years in their paper on grid-connected MG and Diesel generators. [34] use HOMER to model the system. MGs can function well and perform reliably with traditional networks according to [35]. This was revealed in their research on a framework for the techno-economic and reliability analysis of grid-connected MG's.

Material and methods

Data on the climate conditions needed to assess the renewable energy potential at the proposed facility was collected. Again, data on the daily load utilization from the premises has been collected to evaluate the power capacity required from these energy generators. The load data was then validated using the monthly power consumption from the energy meter readings provided by the power utility company. Series of data including the cost of power system equipment, the model, as well as technical specifications of devices were also collected from various distributors and internet sources.

The global solar and wind atlas online tool developed by the World Bank Group was used to assess the energy potential of the site. The geographical coordinates of the location, of 6.1877875, −2.4881826, were inserted into the tool. This is use to determine the wind and solar energy potential at the Sefwi Wiawso Zipline facility. Both reports were generated on the 12th day of October 2021. The specific photovoltaic power output derived at the location was 1385.9 kWh/kWp and the Direct Normal Irradiation (DNI) as well as the Global Horizontal Irradiation (GHI) is 873.3 kWh/m2 and 1743.7 kWh/m2 respectively. Fig. 1 shows the long-term annual PV power output potential at the facility.

The averaging of the annual solar PV power output after the assessment in August proved to be the least. Because it is the peak raining season in Ghana. The month of November provided the highest PV output because it is the onset of the dry season. The average DNI is estimated at 882 kWh/m2/y. Fig. 2 shows the annual and monthly average of the DNI in Wh/m^2. The high levels of solar energy received at the proposed location indicate that a solar PV installation will be viable.

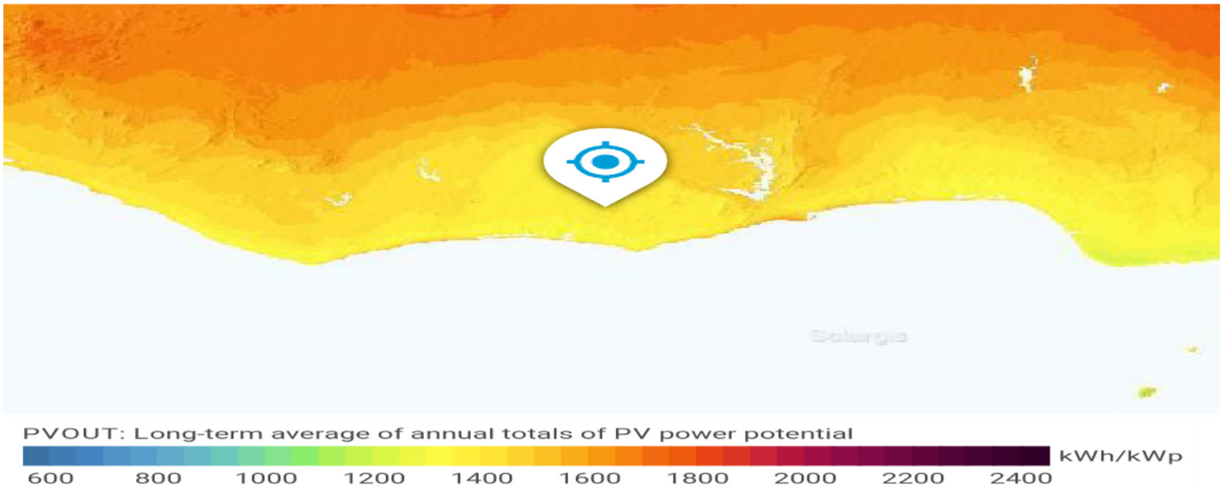


Fig. 2. Photo voltaic power output map of microgrid location (Source: Author).

The wind energy data taken from the global wind atlas was evaluated across a 3 km square diameter. Deploying a wind turbine blade positioned at a height of 50 m above the ground level of the proposed site produced a wind power density of 79 w/m^2 at an average wind speed of 4.17 m/s and wind speed index of 0.9994 m/s. The wind speed index was used to determine the amount of wind registered over an annual cycle. Even though the values did not represent the strongest variation of the wind at the site, they helped to determine the wind variance. The survey conducted across the entire 3 km square diameter indicated that only 14% of the area had wind speeds above 4 m/s. As a result, implementing a wind energy system at the proposed facility would therefore not be cost-effective.

[36] presented a method for assessing the electrical load of a commercial building using the equipment power rating and hours for which the device was utilized within 24 h.

Modeling and simulation of the microgrids are performed to determine the technical specifications required to produce the needed renewable energy for the daily load profile of the facility under study. HOMER is used to perform an optimization of Microgrids to verify the power rating of the energy sources, storage, and power transformation systems with the daily load consumption. Economic analysis is also conducted using the same software [33]. HOMER software helps researchers and engineers to implement effectively the design of a proposed project. The software can carry out various simulations considering different energy sources and the related Balance of Plant (BOP). HOMER poses an embedded algorithm that samples various combinations and options in a single run and groups the sub-systems according to their optimizing variables [37]. The simulation then outputs an optimized design with the proposed least-cost option for the system and an optimum power capacity for each component [34].

A sensitivity analysis is also presented by the software to determine the key variables which when slightly changed but can have a significant effect on the design. Numerous renewable energy technologies have been incorporated into HOMER and data on typical renewable energy sources like Biomass, Micro-hydro, Wind, Solar, etc., are available in the software database. This makes data acquisition for diverse geographical areas less stressful. [31,34] used HOMER software to design, optimize and analyze the performance of a hybrid microgrid system. The solar energy potential and the load profile of the Zipline facility is determined using the software. [34,37,38] carried out a techno-economic analysis of a hybrid microgrid using HOMER.

Load analysis of the facility

The load profile of the proposed facility was therefore determined using [36] method. The power ratings of the equipment and hours of use were used to determine the daily average energy consumed by each device. Data collected for load profiling of the facility was achieved by designing a template where each electrical device together with its corresponding datasheet was entered. Table 1 provides the complete load profile of the facility.

According to [39], a motor load generally cycle on and off between specific times to keep the temperature or pressure at the required limit resulting in the introduction of a duty cycle. The duty cycle of these loads is selected at a factor of 40% of the total operating time. The energy required per device is determined as $E = P * t = kW * h = kWh$. According to the load profile, the proposed nominal daily energy use is 628.71 kWh/d culminating into 18,861.14 kWh/month.

The load profile was validated with the actual energy utilization of the facility using the monthly power consumption recorded by the energy meter installed at the facility. The average from three energy meter bills collected for three months was determined as 16,087.47 kWh/m.

Table 1
Energy utilization on a typical day at the hospital.

Electrical Appliance	Number	Power Rating (kW)	Hours Utilized (h)	Energy Used (kWh/d)
Air Conditioners				
Air Conditioner	10	1.098	4.2	46.116
Charger AC's	8	0.738	4.2	24.7968
Ceiling Cassette AC's	4	5.4	8.4	181.44
Refrigerators				
Domestic Fridge	1	1.8	8.4	15.12
Medical Fridges	4	0.96	8.4	32.256
Medical Freezers	1	0.94	8.4	7.896
Medical Double Door fridge	1	0.58	8.4	4.872
Ultra-cold freezers	1	1.2	8.4	10.08
Deep Freezer	1	0.4	8.4	3.36
Water Dispenser	2	0.6	8.4	10.08
Electrical Motors				
Motor 1	4	5	1.5	30
Motor 2	2	2	1	0.4
Motor 3	1	8.5	1	8.5
Lighting, Fans and Computers				
LED Panel lights	51	0.024	12	14.688
LED Industrial lights	10	0.1	12	12
LED Street Lights	20	0.12	12	28.8
LED Flood lights	6	0.1	12	7.2
LED tube lights	8	0.04	12	3.84
Laptops	10	0.05	5	2.5
Personal Computers	6	0.1	12	7.2
Other Electrical Equipment				
Chargers	12	1.2	12	172.8
Thawer	1	0.96	2	1.92
Bath-room Hand dryer	2	1.8	0.5	1.8
Printers	2	0.01	2	0.04
TV's	2	0.1	5	1
Total Calculated Energy used in a day				**628.7048**
Total Calculated Energy used in a Month				**18,861.144**

Table 2
Hourly consumption of the entire load of the facility.

Hour	1	2	3	4	5	6	7	8	9	10	11	12
Load (kW)	14.55	14.55	14.55	14.55	14.55	21.83	21.83	21.83	40.6	40.6	40.6	40.6
Hour	13	14	15	16	17	18	19	20	21	22	23	24
Load (kW)	37.62	37.62	40.6	40.6	40.6	21.83	21.83	21.83	21.83	14.55	14.55	14.55

The results from the meter reading indicated that the facility utilized its emergency diesel generator for supplying the remaining 2773.67 kWh. The disparity provided by the loads, which operated at different periods during the day necessitated categorizing the hourly consumption of the entire load of the facility as shown in Table 2.

Load specification for the microgrid

The key components of the MG were selected with an emphasis on the proposed architecture. These components included the generating sources, a power converter, an energy storage system, facility loads, the grid, and the MG controller. The evaluation began with inserting the hourly incident load and computed. This was then transformed into monthly and annual load profiles for the facility. The load evaluation yielded an annual average load of 628.65 kWh/d and average power of 26.19 kW. Two sources of conventional power were employed in the model. These are emergency diesel generator and power from the grid. These power sources were incorporated into the design with no additional capital or installation cost incurred since they are present in the current design.

The emergency generator (CAT-C7–158 kW) at the Zipline facility is utilized in the proposed model. A datasheet of various parameters was submitted on the generator by the distributor. The initial capital of the equipment is not used in the model since the device had been purchased before the project proposal. The cost of replacement US$ 26,290.44 as well as operation and maintenance (O&M) were however factored into the model. The minimum load ratio is the minimum load that can be placed on the running generator for its rating was chosen as 20%. Since the generator would not be used as a Combined Heat and Power (CHP) plant, the heat recovery was selected as zero. A minimum runtime per each start of the generator was chosen as 60 min for a lifetime of 90,000 h. The price of diesel in Ghana stood at US$ 1.12 on the 12th day of November 2021. Table 3 presents details of the emergency diesel generator.

Table 3
Engine-generator set data sheet.

Item	GH¢
Initial Capital	GH¢180,000
Replacement Cost	GH¢162,212
O&M Cost (op/hr)	GH¢100
Minimum Load Ratio	20%
Minimum Runtime	60 mins
Lifetime	90,000 H
Price of Diesel	GH¢6.9

An electric grid component is added to the model to serve as a baseload power source. Energy meter readings at the facility indicated that the cost of electricity was US$ 0.22 /kWh for power usage above 500 kWh. The capacity of the electric grid was however chosen as infinite, meaning it can supply the needed load all year round except during scheduled and forced outage periods.

Solar panel modelling, selection and specification for microgrid

The amount of power produced by the solar system is dependent on the size of the panel as well as the solar irradiance and temperature of the selected location. The power output from the PV array was determined using Eq. (5) where parameters such as rated PV array capacity (Ypv), the percentage derating factor (fpv), and the temperature coefficient of power (αP) are known.

$$P_{PV} = Y_{PV}\, f_{PV} \left(\frac{G_T}{G_{T,STC}} \right) [1 + \alpha P (T_c - T_{c,STC})] \tag{5}$$

Where G_T and $G_{T,\ STC}$ are the solar radiation for both current time and standard test conditions respectively. T_c and $T_{c,\ STC}$ are the temperature for the current time and standard test conditions. Since the power required from the PV array must correspond to the rated load of 42 kW, the PV array rating was deduced using Eq. (6).

$$Y_{PV} = \frac{P_{PV}}{f_{PV} \left(\frac{G_T}{G_{T,STC}} \right) [1 + \alpha P (T_c - T_{c,STC})]} \tag{6}$$

The derating factor f_{pv} of the SunPower solar panel was 88%, temperature coefficient of −0.3, T_c and $T_{c,\ STC}$ are 24.67 °C and 25 °C respectively. G_T and $G_{T,\ STC}$ are 0.457 kW/m^2 and 1 kW/m^2 respectively. The rating of the PV array was deduced using Eq. (6) as 94.8 kW. Because a single panel produced 0.335 W, the number of panels required to meet the 94.8 kW is approximately 283 panels. To get uniform distribution of panels and also to make room for future developments, the number of solar panels was estimated at 284, yielding a total power output of 95.14 kW.

Battery energy storage system selection, sizing and financial specification

BESS improves the total capacity of MGs. The successful deployment of a resilient MG must integrate BESS into its design. The power system proposed for the Zipline facility incorporated BESS for two primary reasons. (i) the power system is grid-connected, (ii) the LCOE for solar PV was cheaper than a grid-connected system. The idealized Li-ion battery (Samsung M8068 P2) chosen for the model was selected based on the battery capacity, voltage, and discharge power. It had a nominal voltage of 80.9 V an energy capacity of 5.5 kWh and an Ampere rating of 68 Ah. The battery had a round trip efficiency of 90% with a charge and discharges current of 170 A. The BESS selected has a cycle life of 6000 at a 100% Depth of Discharge (DoD) and 0% minimum state of charge.

The battery energy storage capacity of the system was determined by considering the peak power required by the system in a given period, the DoD, and the roundtrip efficiency. This was achieved by using Eq. (7) below.

$$\text{Battery Capacity (Ah)} = \frac{\text{Power required [kW]} \times \text{Duration required [h]}}{\text{Depth of Discharge [\%]} \times \text{battery efficiency [\%]}} \tag{7}$$

It is realized that the solar energy at the location produced no power output within the hours of 18:00 and 06:00 representing 12 h without power from the solar system. The incident power achieved on the system during the night time was 21.83 kW as found in Table 2. Utilizing a 100% DoD and a roundtrip efficiency of 90%, the M8068 P2 Li-ion battery capacity was evaluated at 291.07 kWh.

Due to the high cost of Li-ion batteries and the tendency of utilizing power from the grid and emergency diesel generator, a capacity factor of 0.8 was explicitly selected for the energy rating of the battery. The estimated battery capacity was calculated at $291 \times 0.8 = 232.9$ kWh. The number of batteries needed to achieve the estimated rating is 42 yielding 231 kWh with the remainder being supplied from the grid. At a voltage of 80.9 V, the batteries were strung in 6 to achieve a total 3-phase voltage of 485 V with 7 strings in parallel.

Even though prices of renewable energy components have fallen over the years, the BESS remains the most expensive single component of an MG system. [40] estimated the cost of Li-ion batteries for utility-scale MGs at $345 /kWh. This was considered in estimating the price of the Samsung M8068 P2 Li-ion battery, which was taken as US$ 345.22 /kWh for the cost of purchase and installation with the same price considered as the cost of replacement. The annual O&M cost of the proposed batteries was US$ 194.49 with a lifetime of 30 years and a throughput of 33,000 kWh.

Inverter selection for the proposed microgrid

A power inverter is required in the MG application to convert the Direct Current produced by the solar panel or energy from the battery storage to Alternating Current which is needed by the electrical appliances available at the Zipline facility for its operation. Selecting the right type of converters for different purposes in a power system tends to influence the optimal performance of the system. After carrying out a quick inspection of the various solar inverters present in the software, the ABB MGS 100 was selected for the design. This solar inverter is capable of combining the activities of all MG components to function as a single unit. It is ideal for the provision of power to commercial facilities connected to an inefficient grid supply. The cost estimation of a bi-directional power inverter carried out by [41] suggested that a typical utility-scale inverter costs $ 105 /kW. This amount was used to determine the cost of the power inverter used in the system. The capital cost of the ABB MGS 100 inverter was inputted into Homer as US$ 104.54 /kW, with a replacement cost of US$ 100.49 and no O&M costs.

Controller selection for grid connected solar pv system

To get all modeled components working as required, producing the optimal power dispatch, and providing the quickest return on investment, a controller was required for the power control strategy of the system. For this project, the dispatch strategy was that the PV operating at full power supplied the primary loads and charged the batteries. The Homer cycle charging controller utilizes a dispatch strategy that causes a connected generator to dispatch full output power to the system whenever there is the availability of supply from the power source. Surplus power goes through predetermined objectives in order of higher to lower priorities including the inter-connecting grid.

Results and discussion

A sensitivity and optimization of the proposed system were carried out using Homer software. The cash flow and economic analysis were conducted considering all the necessary scenarios. The HOMER optimizer presented the optimum combination suitable for implementing the MG for the facility. The simulation was conducted with a project lifetime of 25 years at a nominal discount rate of 13.23% using the load-following dispatch strategy. The optimum design comprised a 94.8 kW of Sunpower X21–335 solar panels and a 231 kWh of Samsung M8068 Li-ion batteries connected to the DC bus, and a 56.31 kW inverter used to convert the DC power from the solar panel and batteries to the AC bus, for the facility. The design also contained a 158-kW diesel generator and an infinite distribution grid connected to the AC busbar.

The COE achieved for the proposed system was US$ 0.14 and an NPC of US$ 515,397.08. The annual operating cost of the proposed system was US$ 28,848.30 with the project expending an initial capital of US$ 45,767.10. The renewable energy fraction from the MG system was 92.7% with the annual fuel consumed by the diesel generator at 225 litres. The generator set operated for 18 h annually producing total energy of 569 kWh. The O&M cost of the generator was US$ 291.73 with only the fuel cost amounting to US$ 252.03. The capital cost of the SPR-X21 solar panel was realized as US$ 25,514.75 and was able to achieve a total energy production of 246,141 kWh/y. The Samsung M8068 battery had an autonomy of 8.82 h with an annual throughput of 79,251 kWh/y. The usable nominal capacity of the battery was attained as 231 kWh. The mean inverter output for the proposed system was 24.3 kW, whilst the mean rectifier output was 0.0111 kW. The total energy purchased from the distribution grid was reduced significantly to 16, 039 kWh.

Cost summary of the proposed microgrid

The annualized cost summary of the proposed project is presented in Fig. 3, which indicated that the total annualized cost of the system is US$ 31,668.40 when all components' costs were added. This comprised the total capital and replacement cost of US$ 2820.10 and US$ 684.28 respectively, an O&M and fuel cost of US$ 28,831.12 and US$ 252.03 respectively, with a salvage gain of US$ 919.12.

The total annualized cost is realized with the component cost analysis. Out of the total cost, the ABB MGS and Samsung M8068 had cost values of US$ 520.58 and US$ 9328.36 respectively., The grid and Sunpower also had cost components of US$ 3457.37 and US$ 18,480.23 respectively, A salvage gain of US$ 118.12 was also made on the CAT-C7 generator after the project life.

Fig. 4 presents the NPC summary of the proposed project. The total NPC of the system was realized as US$ 515,752.32. This comprised the total capital and replacement cost of US$ 45,929.21 and US$ 11,143.09 respectively, an O&M and fuel cost of US$ 466,628.02 and US$ 4103.36 respectively, with a salvage gain of US$ 14,968.71 after the end of the project.

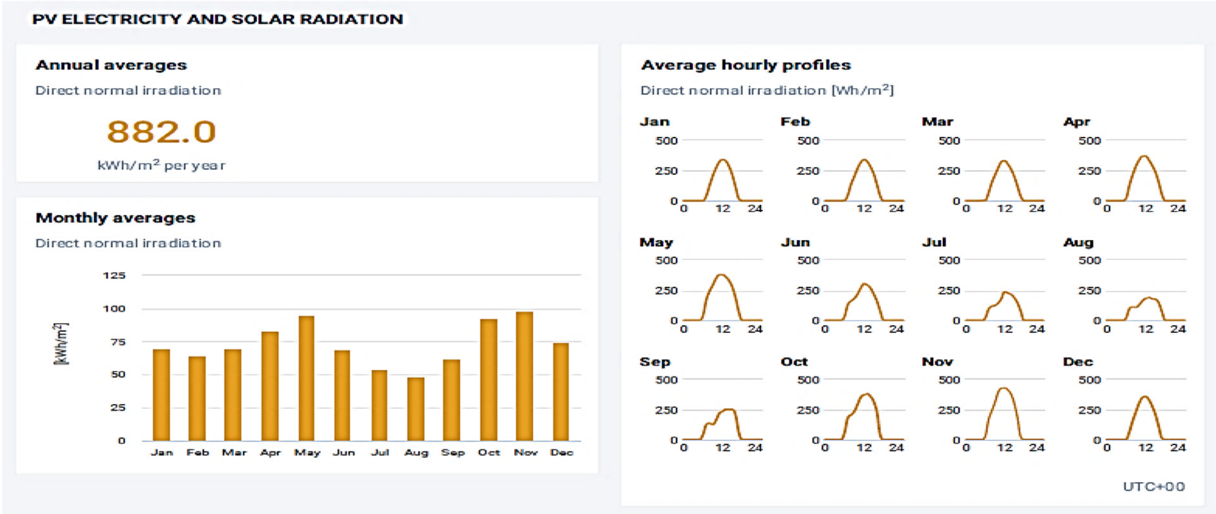


Fig. 3. The annual and monthly average of the DNI in Wh/m2 (Source: Author).

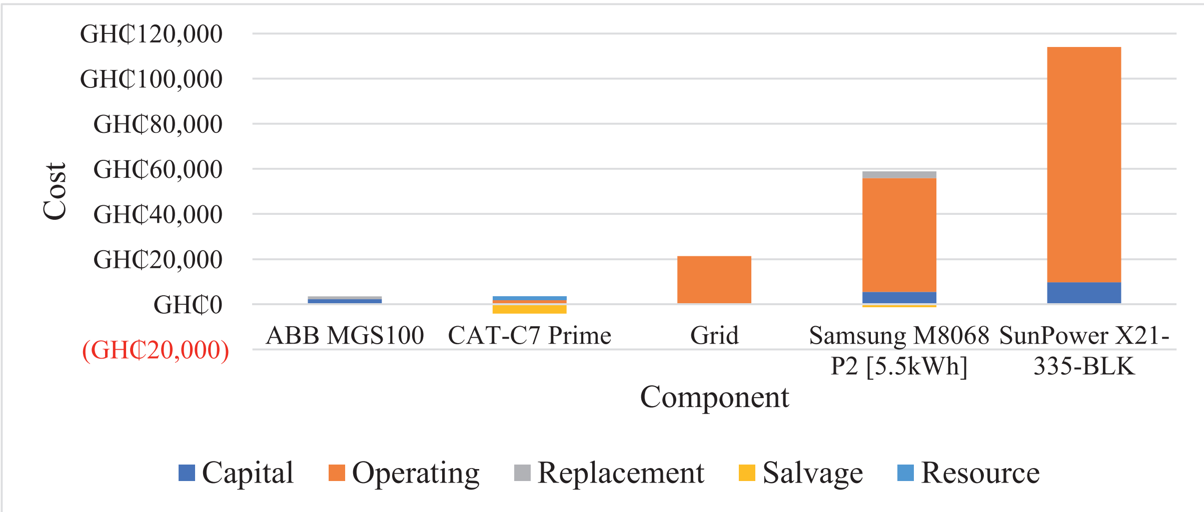


Fig. 4. Annualized cost summary of the proposed project (Source: Author).

Cash flow studies of the proposed microgrid

The cash flow studies comprised the cost type at every stage of the project as well as the annual cash inflow and outflows throughout the project's lifetime. It was also distinguished using nominal or discounted rates where the nominal cash flow has a flat rate incident annually on the project and the discounted rate evaluated the cash flow considering the annual interest rate of 13.23%. The discounted cash flow was employed and analyzed for the project since it tends to depict real-world situations. Fig. 5 shows the discounted cash flow by component and it depicts the total cash flow incurred by each component of the proposed MG for the entire project lifetime.

From Fig. 5, the capital cost was the only cash outflow on the ABB MGS converter. It was estimated at US$ 5915.24 and had a salvage cost being the cash inflow of US$ 782.17 occurring in the final year. The CAT-C& diesel generator had a cash outflow occurring within the first and final year of the project starting at US$ 524.80 at a discount rate of 13.23%. It also had an end-of-project salvage rate of US$ 10,778.28. Purchases made from the grid within the project time frame started at US$ 3336.95 in year 1 at a nominal discount rate.

The Samsung M8068 batteries had a capital cost of US$ 14,499.19 occurring in year 0 and an annual discounted cash flow of US$ 7883.95 from year 1 to 25. The component had a salvage cash inflow of US$ 3408.27 at the end of the project. The Sunpower X21–335 PV had a capital cost of US$ 25,514.75 and an annual discounted cash outflow of 100,720 from years 1 to 25.

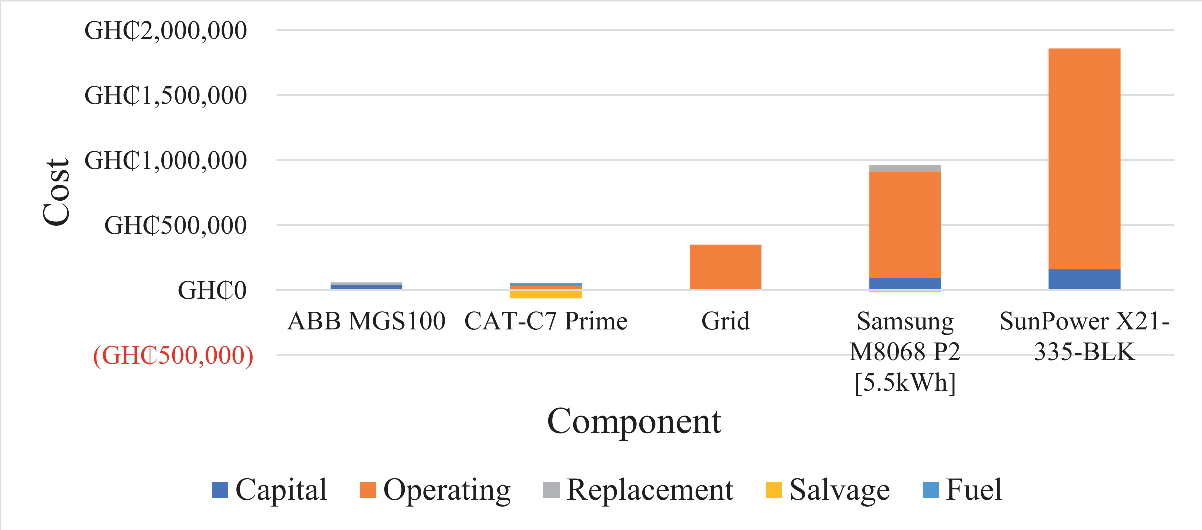


Fig. 5. Net present cost summary of proposed system (Source: Author).

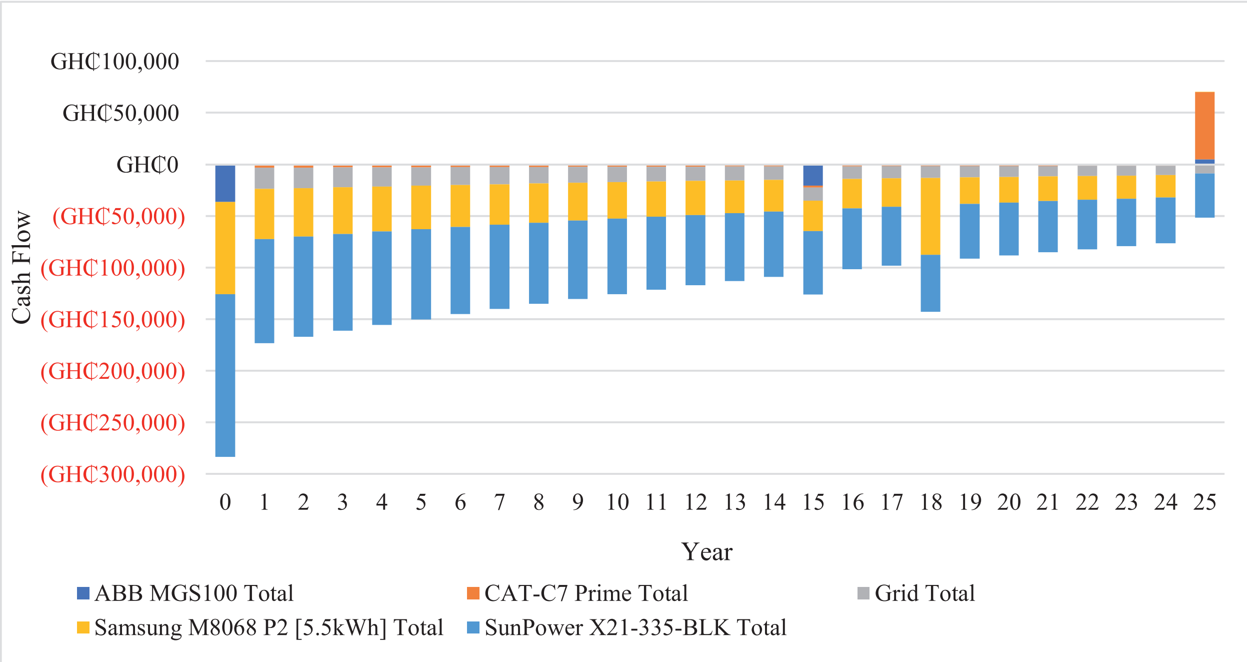


Fig. 6. Discounted Cashflow by Component (Source: Author).

Fig. 6 shows the discounted cash flow categorized by its cost component. It indicates that a capital cost outflow of US$ 45,929.17 occur in year 0, with a salvage cost inflow of US$ 14,968.72 gained in the final year of the project. A replacement cost outflow of US$ 3345.38 and US$ 7797.73 was expended on the project during the 15th and 18th years of the project respectively. This was due to the purchase of a new ABB MGS inverter and Samsung M8068 battery pack. An annual cost of US$ 27,826.58 and US$ 243.11 at a nominal discount rate of 13.23% was spent on the project due to O&M and the purchase of fuel for the CAT-C7 generator set.

Economic comparison of the current and proposed system

An economic evaluation of the work was achieved when the proposed system was evaluated against the existing system. The proposed system is made up of a grid, a diesel generator, solar PV, batteries, an inverter, and an MG controller. This is

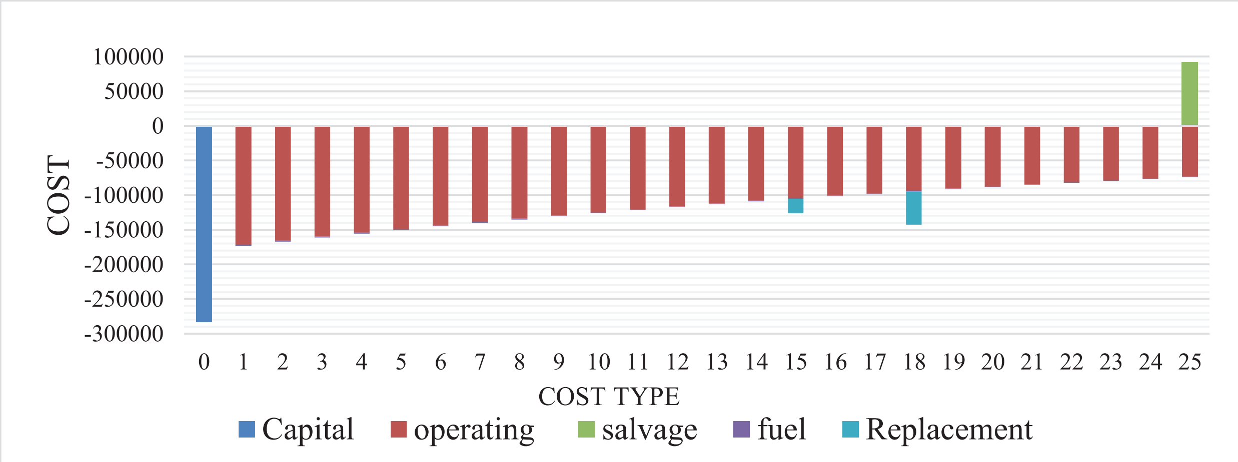
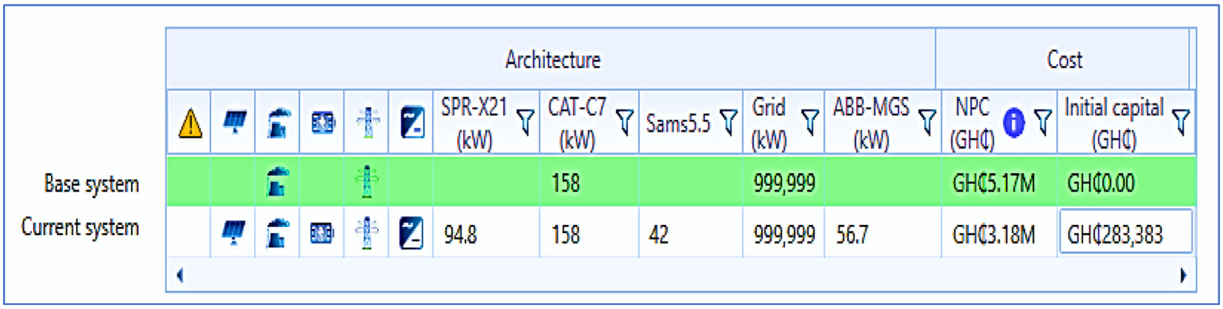


Fig. 7. Discounted cashflow categorised by their cost type (Source: Author).



	Architecture											Cost	
							SPR-X21 (kW)	CAT-C7 (kW)	Sams5.5	Grid (kW)	ABB-MGS (kW)	NPC (GH₵)	Initial capital (GH₵)
Base system								158		999,999		GH₵5.17M	GH₵0.00
Current system							94.8	158	42	999,999	56.7	GH₵3.18M	GH₵283,383

Fig. 8. Architecture of the base system and current system (Source: Author).

compared to the existing system, which is made up of a grid and a diesel generator. The existing system was chosen as a base case and compared to the proposed architecture. Fig. 7 shows the architecture of the two systems to be compared.

The NPC of the base case was US$ 826,580.23, whereas that of the proposed case is US$ 515,397.08. The base case has no initial capital however the proposed case has an initial capital of US$ 45,929.17. The present worth of the project is US$ 322,563.21, with an annual worth of US$ 19,806.16. The RoI of the project is estimated at 45% with an IRR of 50.1%. A simple payback period of 1.99 years and a discounted payback of 2.11 years were achieved by implementing the proposed system. Fig. 8 shows the annual cash outflows for the two systems. The difference in cash flow was derived by subtracting the base case system from the proposed system. Except for year 0 where the capital cost of the project caused an excess expenditure of US$ 45,767.10. It is realized that when the proposed system is implemented, an annual gain of over US$ 19,465.15 will be made each year causing the facility to break even and start making profits after the second year.

Electrical characteristics of the designed microgrid

The designed MG sought to develop additional means of providing electrical power to a commercial facility in Ghana. The results indicated a total electricity production of 262,749 kWh/y. The Sunpower X21–335 Solar panel produced 246,141 kWh/y representing 93.7% of the total electricity produced. The CAT-C7 diesel generator produced 569 kWh/y (0.216%) and 16,031 kWh (6.10%) of electricity was purchased from the Grid. The system was primarily made up of 229,457 kWh/y of AC load. Excess electricity of 13,844 kWh/y representing 5.27% of total generated power is derived from the system. The electricity load was met with no shortage capacity. The system had a renewable energy penetration of 92.8%. Fig. 9 shows the monthly average electricity production indicating that for most of the year, solar power was available as the main source of energy and was also the cheapest. It had little capacity shortage, and on those occasions, power from the distribution grid or diesel generator supported the load requirement. The high production of the SPR-X21 was a result of the solar energy feeding the load and charging the batteries.

Technical analysis of the CAT-C7 emergency diesel generator

The diesel generator was placed below in the order of priority being the last resort for energy generation for the facility. It only operated when there was no supply from the solar panels, battery storage, and distribution grid. The diesel

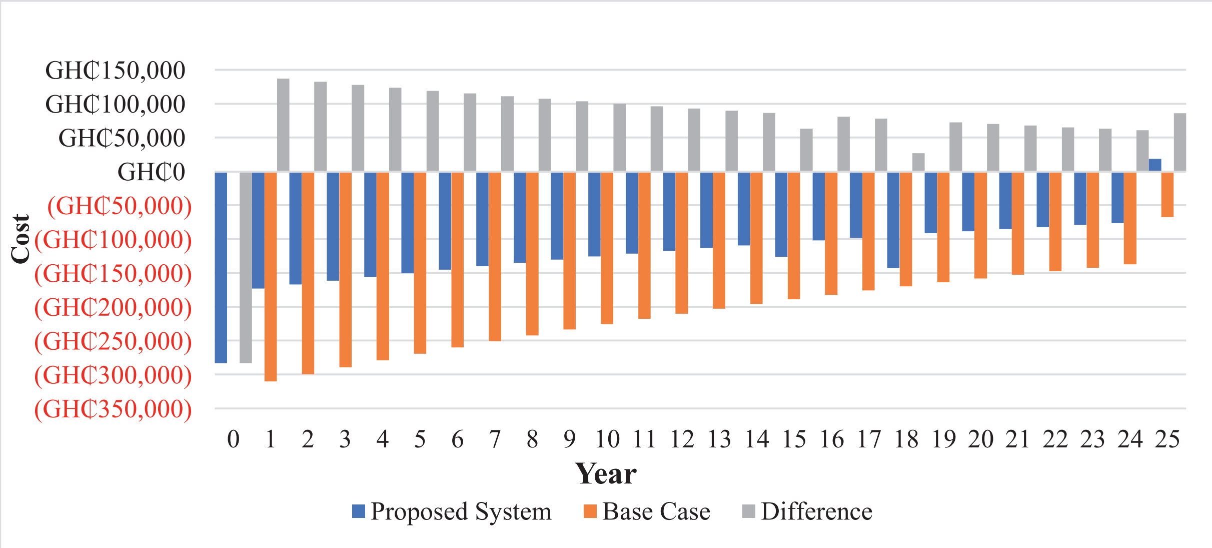


Fig. 9. Cost comparison of base case and proposed case (Source: Author).

generator operated for 18hr/y with 14 starts. The operational life was estimated at 5000 yr at a capacity factor of 0.0411%. Operating the CAT-C7 diesel generator yielded a fixed generation cost of US$ 20.42 /hr and a marginal generation cost of US$ 0.32 /kWh. The electrical output of the generator was 31.6 kW and consumed 225 L, with specific fuel consumption of 0.396 L/kWh. The fuel/energy input was 2217 kWh/y and a mean electrical efficiency of 25.7%.

Technical analysis of the Samsung M8068 battery storage

The second priority of electricity supply to the load was placed on the batteries. It had 42 single units, each at a capacity of 5.5 kWh and a string size of 6 batteries per string. 7 of such strings were connected in parallel to deliver a three-phase voltage of about 420 V. The batteries had an autonomy of 8.82 hr and a storage wear cost of US$ 0.01/kWh. The nominal capacity of the batteries was 231 kWh, a lifetime throughput of 1386,000 kWh, and an expected life cycle of 17.5y. The average energy stored by the device was 83,419 kWh/y, whilst the energy output was 75,185 kWh/y. A storage depletion of 113 kWh/y was recorded on the batteries, which accommodated energy losses of 8348 kWh/y at an annual throughput of 79,251 kWh/y. Fig. 10 shows the monthly state of charge of the batteries. It has been realized that the batteries assumed a 100% state of charge between the hours of 12:30 and 18:00 and 0% between 5:30 and 7:00. If there was no power from the solar panel at that time, then either the grid or the diesel generator came in to support the load.

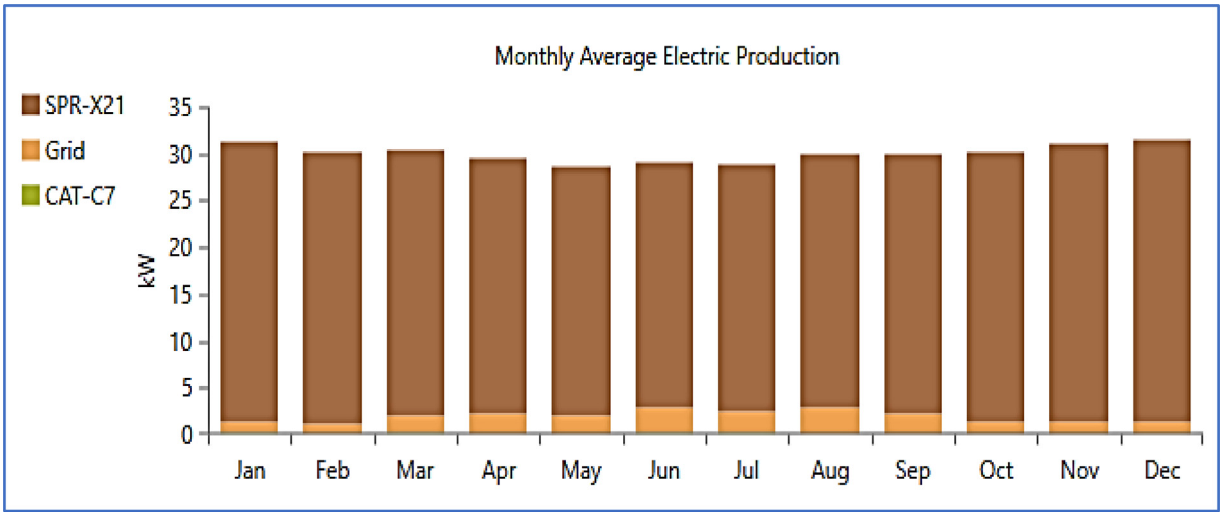


Fig. 10. Monthly average electricity production (Source: Author).

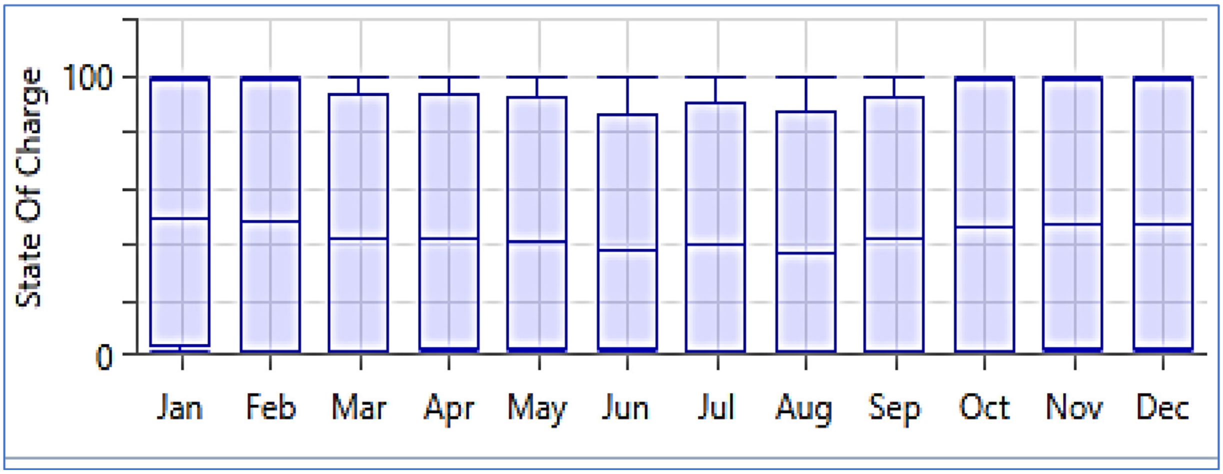


Fig. 11. Monthly state of charge of samsung battery storage (Source: Author).

Fig. 10 indicates that a 100% state of charge was realized only between October to February, whilst for the remaining 7 months, the batteries could not assume full charge due to solar energy shortfall from March to September. Other generators would therefore be needed during these periods.

Technical analysis of the Sunpower X21–335-BLK solar PV

The main source of electricity generation to the facility was delivered by the solar panels producing about 93.7% of the total power supplied annually. It had a rated capacity of 94.8 kW and a mean output power of 28.1 kW. The mean output energy produced by the solar panels is 246,141 kWh/y. It had a maximum output of 107 kW and a minimum output of 0 kW during periods of no sunshine. A PV penetration of 107% was recorded by the system with an annual operational time of 4356 hr/y and an LCOE of US$ 0.08 /kWh. Fig. 11 shows the power output of the Sunpower X21 solar panels. The operational period of the device was between 6:00 to 18:00. Electrical power was supplied to the load from 6:00 until 10:00 when excess electricity was produced by the panels. The excess electricity was used to charge the Li-ion battery.

At 16:00, when solar energy was minimal, the little electricity produced was used to supply only the load until no energy was produced making the system invoke the energy supply of the batteries. The whole cycle began the next morning and continued throughout the year. During certain times of the year with low battery storage, either the grid or diesel generators supported power production.

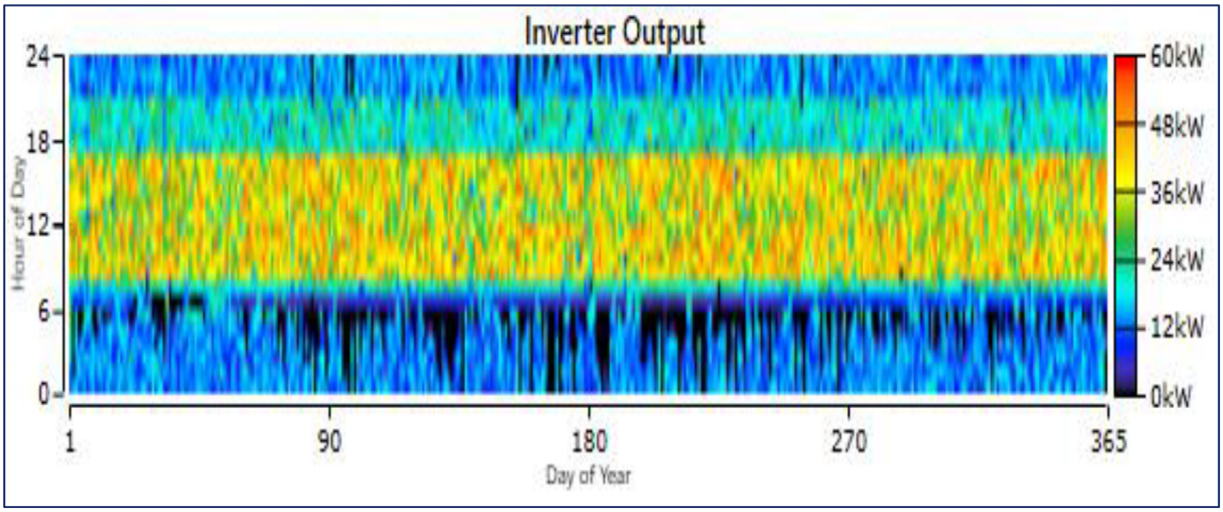


Fig. 12. Output power of the ABB MGS inverter (Source: Author).

Table 4
Energy purchase and charges for both current and proposed architecture.

	Current Architecture		Proposed Architecture	
	Energy Purchased (kWh/m)	Energy Charge (GH¢)	Energy Purchased (kWh/m)	Energy Charge (GH¢)
January	18,936.20	25,185.15	829.35	1103.04
February	16,704.69	22,217.24	694.49	923.67
March	19,883.89	26,445.58	1439.96	1915.14
April	18,730.49	24,911.55	1554.18	2067.05
May	18,780.19	24,977.65	1376.89	1831.27
June	18,863.22	25,088.08	1910.93	2541.54
July	19,080.30	25,376.80	1670.79	2222.16
August	20,138.93	26,784.77	2065.87	2747.61
September	18,953.97	25,208.78	1544.34	2053.98
October	19,077.60	25,373.21	1008.68	1341.54
November	18,439.66	24,524.75	964.84	1283.24
December	19,174.17	25,501.65	978.86	1301.88
Average	**18,896.94**	**25,132.93**	**1336.60**	**1777.68**

Technical analysis of the ABB MGS converter

The ABB MGS Converter worked all year round to convert the DC supply from the solar panel and batteries to AC for the electrical load. The total capacity of the inverter is 56.7 kW. The mean output power of the inverter was 24.3 kW. The capacity factor for both inverter and rectifier were 42.9% and 0.0322% respectively with their hours of operation at 8277 hr/y and 9 hr/y. The input and output energies were 212,951 kWh and 224,159 kWh respectively for the inverter. The maximum output power is 56.7 kW as shown in Figure.

Technical analysis of the distribution grid on the microgrid system

Even though the cost of electricity was high when using the grid, its inclusion in the MG design was important to improve the overall reliability of the electricity supply to the facility. The grid operated intermittently when there was no supply from the solar panels and storage battery. The energy purchased from the grid was between 694 kWh/m and 1911 kWh/m, whilst the energy charge ranged between US$ 149.70 /m and US$ 411.92/m.

Table 4 below shows the energy consumed and electricity bills paid for both the current and proposed architecture. It was realized that the current system had an average monthly energy purchase of 18,897 kWh/m and an energy charge of US$ 4073.42. However, after implementing the proposed strategy, the energy purchased from the grid got reduced to 1337 kWh/m with an average monthly energy charge of US$ 288.17.

Conclusion and recommendations

The study focused on providing a cheaper and more reliable power supply to a commercial facility in Ghana using renewable energy sources and components. The work, therefore, proposed a design of a robust MG capable of meeting the electrical load demand of the Zipline drone delivery service in Sefwi-Wiawso. The following are the findings. The load profile for the facility has a daily energy utilization of 628.70 kWh resulting in monthly energy consumption of 18.86 MWh and an annual production of 223.32 kWh. The monthly energy purchased for the current system is 18,897 kWh/m with an energy charge of US$ 4073.42. The proposed system however had an energy purchase of 1337 kWh/m with an average monthly energy charge of US$ 288.17.

Solar PV is identified as the viable energy resource to achieve the desired power for the proposed MG. The optimum design achieved from the HOMER optimizer comprised 94.8 kW Sunpower solar panels, 231 kWh Samsung M8068 Li-ion batteries, 56.31 kW inverter, 158 kW diesel generator, and an infinite distribution grid supplying an average load of 628.65 kWh/d.

The system produces a total electricity of 262,749 kWh/y. This energy was capable of supplying an AC load of 229,457 kWh/y with a system energy loss of 19,448 kWh/y and excess electricity production of 19,448 kWh/y. The produced energy was made up of 93.7% solar energy, 0.2% diesel engine generation, and 6.1% grid purchases. This showed a high renewable energy penetration, indicating reduced greenhouse gas emissions and improved environmental conditions when the proposed system is implemented.

Economically, the proposed system had an initial financial burden of US$ 45,929.17 and an NPC of US$ 515,752.35 as compared to the current system with no initial capital but has an NPC of US$ 838,315.56 considering a project lifetime of 25 years. The designed MG had a present worth of US$ 322,563.21, an annual worth of US$ 19,806.16, a 45% RoI, and a discounted payback period of 2.11 years after which the facility starts making gains.

The paper recommends the following:

- Renewable energy technologies in the country should be implemented for large-scale electricity production to reduce the high cost of electricity.
- Feed-in tariff policies should be implemented in the distribution system. This will encourage commercial facilities to invest in the technology to sell excess power to the grid.
- Mortgage strategies should be designed by financial institutions to support commercial entities to venture into individual power production using renewable energy sources.

The study uncovered the high cost of electricity from conventional sources. This was confirmed at the Zipline facility where the monthly energy consumption of 18.86 MWh cost about US$ 4073.42. An optimum MG was successfully designed using Solar PV, with the capacity of providing about 93.7% of renewable electrical energy to power the commercial facility. Further research focusing on designing an MG system using a Load-Following Controller (LFC) strategy tied to a distribution system with an effective feed-in tariff should be explored. Fig. 12

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

David Mensah Sackey: Conceptualization, Writing – original draft, Visualization, Investigation, Writing – review & editing. **Michael Amoah:** Methodology, Formal analysis, Investigation. **Adam B. Jehuri:** Supervision, Project administration, Validation. **De-Graft Owusu-Manu:** Supervision, Writing – review & editing. **Amevi Acapkovi:** Formal analysis, Writing – review & editing.

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